Waves at boundaries

antocciphysics

UNIT 6 / DAY 4 / HONORS

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Describe reflection and transmission at a boundary.

SUCCESS CRITERIA - I can...

1. Identify incident, reflected and transmitted waves.
2. Distinguish fixed-end from free-end pulse reflection.
3. Track energy without assuming equal amplitudes.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

Echoes return at **0.003 s** and **0.005 s** with the same wave speed.
1. **Compare ranges**.
2. Does delay alone find mass?

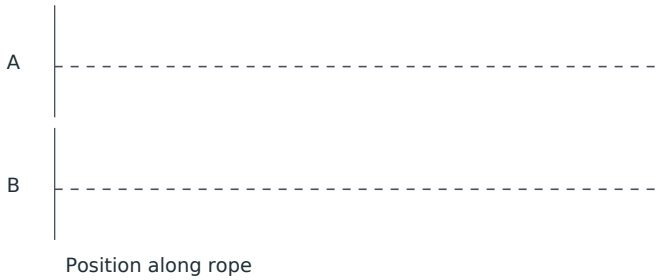
Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $f_{reflected} = f_{incident}$.
At a stationary boundary. A displacement pulse returns inverted from an ideal fixed end and upright from an ideal free end.

Worked reasoning: At a fixed end, an upward rope pulse reflects downward. At an ideal free end it reflects without inversion. The end of the rope is held differently in these two cases.

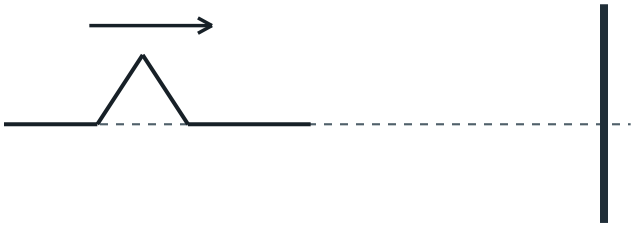
Your turn: Draw incident and reflected pulse snapshots for both ends; **show propagation directions**.



Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

An upward rope pulse reaches an ideal fixed end. **Sketch the reflected pulse** and its travel direction.



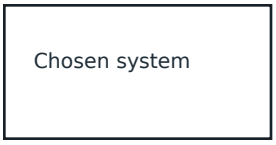
Sketch and label your response.

Marzano check - circle: ① ② ③

04 Apply the physics

A pulse carries **0.80 J** to a boundary. Of this, **0.20 J** is reflected and **0.10 J** is absorbed. **Complete the energy-flow diagram**. **Find transmitted energy** and the reflected, absorbed and transmitted fractions.

Draw and label all energy transfers across the boundary.



Stored energy change: _____ J

Marzano check - circle: ① ② ③

Read down the left column, then down the right. Use the open space to draw, calculate and explain.

Waves at boundaries

antocciphysics

Continue down each column.

05 Explain your reasoning

Does a smaller transmitted amplitude prove frequency changed? **Explain what usually fixes the** frequency at a stationary boundary.

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A **0.4 kg** vapor releases **120000 J** while condensing at constant temperature. **Find latent heat** and signed Q for the sample.
Label temperatures, samples and relevant energy transfers.

Sample A

Sample B

Label temperatures and draw energy-transfer arrows.

Marzano check - circle: 1 2 3

07 Exit check

At a stationary boundary frequency stays **5 Hz** while wave speed changes from **10** to **20 m/s**. **Find wavelength** on each side.



Position along rope

Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

A pulse reflects from a fixed end and meets an equal incident pulse. At an instant their displacements are **+2** and **-2 cm** at one point. **Find total displacement** and **explain why zero there** does not mean all energy vanished.



Sketch and label your response.

Marzano check - circle: 1 2 3